Published in final edited form as:
Nat Methods. 2024 April ; 21(4): 657–665. doi:10.1038/s41592-024-02203-y.

Bright and stable monomeric green fluorescent protein derived from StayGold

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Author contributions

G.D.L., O.M.S., and F.V.S. developed mBaoJin and characterized it in vitro. F.V.S. and T.P.K. obtained mutants of mBaoJin and characterized their properties in vitro. A.V.V. and H.Z. purified mBaoJin. H.Z., K.D.P., and F.V.S. characterized mBaoJin in mammalian cells and performed live cell imaging. M.D. measured QY of proteins. S.P. and H.Z. performed OSER assay. H.Z. characterized proteins in vivo. H.Z. and W.Z. perfromed ExM. L.C., X.Y., H.Z., and K.D.P. performed SIM. M.M.P., A.S.G. and A.S.M. performed BaLM microscopy. A.G., V.B. and W.Q. performed X-ray data collection. A.Yu.N. performed crystallization. V.R.S. perform crystallization, structure solution, refinement and analysis. K.D.P., F.V.S, H.Z., M.M.P., V.R.S., and A.S.M. wrote the manuscript. K.D.P. and F.V.S. supervised all aspects of the project. All authors reviewed the manuscript.

Competing interests

K.D.P. is the co-founder of a company that pursues commercial applications of expansion microscopy and is listed as an inventor on several patent applications concerning the development of new expansion microscopy methods. All other authors have no competing interests.

Code availability

Custom code used in this study for BaLM imaging reconstruction is available at https://github.com/Perfus/mBaoJin.

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Abstract

The high brightness and photostability of the green fluorescent protein StayGold make it a particularly attractive probe for long-term live cell imaging. However, its dimeric nature precludes its application as a fluorescent tag for some proteins. Here, we report the development and crystal structures of a monomeric variant of StayGold, named mBaoJin, which preserves the beneficial properties of its precursor while serving as a tag for structural proteins and membranes. Systematic benchmarking of mBaoJin against popular GFPs and other recently introduced monomeric and pseudomonomeric derivatives of StayGold established mBaoJin as bright and photostable FP, exhibiting rapid maturation and high pH/chemical stability. mBaoJin was also demonstrated for super-resolution long-term live cell imaging and expansion microscopy. We further showed the applicability of mBaoJin for neuronal labeling in model organisms, including *Caenorhabditis elegans* and mice.

Editor’s summary

mBaoJin is a monomeric derivative of the bright and photostable green fluorescent protein StayGold. mBaoJin offers favorable photophysical properties for use in diverse protein tagging and subcellular labeling applications.

Introduction

Fluorescent proteins are valuable molecular probes for live cell imaging and super-resolution microscopy that enable specific labeling of target proteins, cellular structures, and organelles[1]. Fluorescence brightness and photostability are two major traits that define the overall utility of an FP for fluorescence microscopy. Therefore, since cloning the first GFP in 1992, protein engineers have been putting much effort and thought into improving the brightness and photostability of FPs, which are crucial factors for extending the spatial and temporal scales of fluorescence imaging. Serendipitously, recently discovered GFP StayGold (a point mutant of wild-type GFP from *C. uchidae*) exhibits exceptional photostability of an order of magnitude higher than that for any currently available FP while characterized by high molecular brightness[2]. However, StayGold forms an obligate dimer that limits its applicability as a fluorescent tag for protein and membrane labeling due to potential mislocalization of fusions and constraints on biosensor development. To address this issue, we employed a directed molecular evolution approach to generate a monomeric version of StayGold, referred to as mBaoJin (‘bao jin’ means ‘stay gold’ in Mandarin), which retains the beneficial properties of its precursor. We solved the crystal structures of mBaoJin under different pH and compared them to those of StayGold and other popular GFPs to reveal monomerization mutations and elucidate molecular determinants of its high photostability. High intracellular brightness and photostability of mBaoJin, in combination with the exceptional chemical stability, enabled its application for long-term super-resolution imaging of subcellular dynamics and expansion microscopy (ExM) of cells and tissues.

Results

Monomerization of StayGold

To facilitate screening for monomeric StayGold variants in *E. coli*, we took advantage of the finding that the DNA-binding domain of the AraC protein ($AraC_{DNA}$), when not connected to a dimerization domain was unable to activate detectable transcription from the wild-type pBAD promoter, but $AraC_{DNA}$ protein in fusion with dimeric protein activated pBAD promoter in bacteria[3]. To test this possibility, we constructed an expression vector, which had fusion of FP of interest with $AraC_{DNA}$ domain under the control of the rhamnose promoter and a reporter gene, mTagBFP, under the control of the arabinose promoter (Fig. 1a). According to our model, dimerization of $AraC_{DNA}$ promoted by a dimeric FP of interest drives a higher expression level of reporter gene compared to monomeric FP (Fig. 1b). Indeed, validation of the expression vector using FPLC revealed that more monomeric StayGold variants induced lower mTagBFP expression compared to more dimeric StayGold mutants (Fig. 1c,d) having various mutations (Supplementary Fig. 1). Correspondingly, during each round of random mutagenesis we selected the brightest green colonies possessing dimmer blue fluorescence confirming their oligomeric state with FPLC. After eight rounds of directed evolution, we selected a bright monomeric variant, which was further modified with the N- and C-termini from mNeonGreen protein to ensure its stability in mammalian cells, resulting in a GFP named mBaoJin that had the S55T, H77R, E80G, Q140P, H141Q, C165Y, N171Y, and T201A mutations (Supplementary Fig. 2 and 3). FPLC demonstrated that mBaoJin was ~99% monomeric at a high concentration (56 μM), which was higher than monomeric fraction for mNeonGreen (95% monomeric). At the same time, parental StayGold did not exhibit a monomeric fraction (Fig. 1e). To validate the monomeric behavior of mBaoJin in mammalian cells, we expressed its fusion with CytERM in live HeLa cells and evaluated the fraction of healthy transfected cells without visible OSER whorl structures.[4] For comparison, we used the tandem and monomerized versions of StayGold (td8oxStayGold[5], StayGold-E138D[6], and mStayGold (a.k.a. QC2-6 FIQ)[5]) as well as tried-and-true monomers (mEGFP, mClover3, mGreenLantern) and weak dimers (EGFP, Venus). mBaoJin scored 93.7%, reaching the threshold of monomericity in OSER assay[7] (Fig. 1f, see Supplementary Fig. 4 for more representative images, see Supplementary Table 1 for complete statistics). These results established mBaoJin as a monomeric protein.

Crystal structure of mBaoJin

To understand the role of the introduced mutations, we analyzed the deposited X-ray structures of the StayGold dimer[6] and solved X-ray structures of mBaoJin with 1.45, 1.8, and 1.6 Å resolution at pH 6.5, 4.6 and 8.5, respectively (Fig. 2a, and Supplementary Fig. 5). Two mutations (S55T and H141Q) were internal to the β-can, and six others were external (Supplementary Fig. 3). Among external mutations, Q140P mutation can potentially disrupt the dimeric interface of StayGold observed in X-ray structure since Q140 residues from both subunits form symmetrical contacts with N137 and T153 residues from opposite subunits (Supplementary Fig. 6). Analysis of FPLC chromatograms (Fig. 1d) and mutations introduced during StayGold evolution (Supplementary Fig. 1) revealed that the C165Y mutation introduced in the 4th round of directed evolution resulted in efficient monomerization of the protein. Probably, bulky tyrosine residue in position 165 sterically prevented the efficient formation of the dimer by unfavorable interactions with, *e.g.*, R144. Analysis of the mBaoJinstructure at pH6.5 confirmed the presence of fewer contacts between A and B subunits at the dimeric interface of mBaoJin compared to StayGold (Supplementary Table 2) and revealed chromophore surrounding (Fig. 2b). The differences between mBaoJin structures at different pH were insignificant (average r.m.s.d values between Cα-atoms of mBaoJin at pH 6.5 and 4.6, mBaoJin at pH 6.5 and 8.5 and mBaoJin at pH 4.6 and 8.5 were 0.46, 0.12 and 0.47 Å, respectively). The chromophore of mBaoJin was more planar at acidic pH values (Supplementary Table 3 and Supplementary Fig. 7). Chloride pocket near chromophore was found in all structures of mBaoJin (Fig. 2c). The chromophore environment and chloride binding pocket in mBaoJin were similar to those in StayGold[6] (Supplementary Fig. 8). The molecular brightness of mBaoJin was practically unaffected after reverting the internal amino acid T55 to the original one, though this mutation resulted in a 1.5-fold increase in the corrected photostability[8] of mBaoJin (Supplementary Table 4 and Supplementary Fig. 9; T55S mutation was not included into the final variant as it was discovered in the final stage of the study when full characterization of mBaoJin was completed). To further understand the molecular basis of enhanced photostability of mBaoJin, we performed structure-guide mutagenesis of amino acids at positions 134, 137, 152 (enumeration follows that in PDB) interacting with phenolic group of the chromophore (Fig. 2b). We discovered that the mBaoJin/N137K, mBaoJin/S134P and mBaoJin/V152E variants photobleached 1.5- 6.5- and 4.5-fold faster than mBaoJin (Supplementary Fig. 10 and Supplementary Table 5). The comparison of the chromophore environment of the photostable mBaoJin protein and the less photostable mNeonGreen and EGFP proteins revealed a more rigid chromophore fixation in the case of photostable mBaoJin protein due to the formation of more hydrogen bonds and hydrophobic contacts (Supplementary Fig. 11).

Characterization of mBaoJin in solution and cultured cells

Having validated mBaoJin as monomeric, we further characterized its spectral and biochemical properties in solution (Fig. 2d,e, Supplementary Fig. 12, 13, 14, and Supplementary Table 6). The introduced mutations did not alter absorption and fluorescence spectral profiles (Supplementary Fig. 12a), however, mBaoJin had 1.09-fold lower molecular brightness than StayGold though it was 1.24-fold brighter than mNeonGreen (Supplementary Table 6). At low excitation intensity (~13 mW/mm^2, typical for live cell imaging), mBaoJin photostability was 15-, 9-, and 130-fold higher than that for EGFP, mNeonGreen, and mGreenLantern, respectively, although this difference was less substantial at the high power (~120 mW/mm^2; Supplementary Table 6 and Supplementary Fig. 12). In turn, mStayGold was 1.4-, 1.8-, and 3.5-fold more photostable than mBaoJin at low (~13 mW/mm^2), medium (~59 mW/mm^2), and high (~120 mW/mm^2) power, respectively (Supplementary Table 6 and Supplementary Fig. 12). As compared to mBaoJin, the photobleaching rates of EGFP, mNeonGreen, and mStayGold, in solution under 18 W/mm^2 light power, which corresponds to the highest illumination intensity we used in BaLM imaging, were the same, except 2.1-fold lower photostability for EGFP (Supplementary Fig. 12). Rapidly maturing FPs are beneficial for imaging gene expression dynamics and possess higher brightness in growing cells than slowly maturing FPs[9]. mBaoJin exhibited rapid maturation at 37°C with a maturation half-time of 7.5 min, which was 1.8-, 2.2-, and 1.8-fold faster than those for StayGold, mNeonGreen, and EGFP, respectively (Fig. 2d, Supplementary Table 6), making mBaoJin one of the fastest maturing FPs[9]. Furthermore, mBaoJin fluorescence appeared to be highly chemically stable. For example, compared to other monomeric GFPs, mBaoJin was characterized by high pH stability with pK_a value of 4.37 (Fig. 2e), which shifted by −0.19 pH units in the presence of 300 mM sodium chloride (Supplementary Table 7 and Supplementary Fig. 15); the pK_a value of mBaoJin was less susceptible to the presence of NaCl as compared to StayGold, mNeonGreen, and EYFP, which changed their pK_a values in the presence of chloride ions by −0.62, −0.36 and +1.55 pH units, respectively. The apparent K_d value of chloride ion binding to mBaoJin was about 3 mM (Supplementary Fig. 16). Brightness and photostability of mBaoJin increased in the presence of 140-300 mM chloride anions by 15 ± 4 % and 2.24 ± 0.03 - fold, respectively (Supplementary Fig. 16, 17). The high pH stability of mBaoJin prompted us to check its chemical stability by extended incubation with 6M GndHCl (guanidinium hydrochloride). After incubation of mBaoJin in 6M GdnHCl for 24 h it increased its fluorescence by 18%, while hfYFP, one of the most chemically stable FPs[10], increased its fluorescence by 1%, StayGold lost 2% of its fluorescence, and mNeonGreen fluorescence was completely quenched (Supplementary Fig. 12l).

Next, it was essential to verify that the introduced mutation did not have a detrimental effect on mBaoJin brightness and photostability in mammalian cells. For this, we co-expressed mBaoJin with red FPs (FusionRed or mCherry) via P2A self-cleavage peptide in two possible orientations in HEK and HeLa cells and used red fluorescence for expression level normalization. For comparison, we used other popular/bright GFPs (EGFP, mGreenLantern, mNeonGreen, and AausFP1) as well as parental StayGold and its tandem and monomeric versions; we chose to use StayGold and StayGold-E138D[6] appended with the N-terminal region of EGFP (referred to as n1) and ten amino acids at the C-terminus of dfGFP (ref.[11]) (refered to as c4) as described in the original StayGold publication[2] since they exhibited consistently improved intracellular brightness (Supplementary Fig. 18, 19 and Supplementary Table 8). Compared to EGFP, the intracellular brightness of the mBaoJin was consistently higher by ~70-140% while comparable to that of (n1)StayGold(c4) and mStayGold in live HEK and HeLa cells (Fig. 2f,g and Supplementary Table 8). PFA-fixation did not affect mBaoJin brightness, in turn fluorescence of (n1)StayGold(c4) and (n1)StayGold-E138D dropped by 3.0- and 2.4-fold, respectively (Supplementary Fig. 20). Furthermore, high chemical stability of mBaoJin and its resistance to PFA fixation were associated with 17-, 1.4-, 3.8-fold higher brightness in HEK cells processed using the modified Expansion Microscopy protocol compared to mNeonGreen, hfYFP, and mStayGold, respectively (Fig. 2h), enabling multicolor 3D super-resolution imaging of cultured cells (Supplementary Fig. 21, Supplementary Video 1).

For photobleaching experiments in mammalian cells, we used 3.5-fold higher illumination power (*i.e.*, 50.5 mW/mm^2), which we typically use for live cell imaging, in order to observe a difference in photobleaching rate within shorter periods. In live HEK cells, mBaoJin photobleached 4.6-85-fold slower than other established GFPs, although StayGold and its other derivatives exhibited 1.9-2.3-fold higher photostability (Fig. 2i and Supplementary Table 8). The tandem dimeric version of StayGold, td8ox2StayGold, outperformed all tested GFPs in terms of brightness and photostability in live HEK cells and given its good localization in fusion with certain structural proteins including clathrin, ensconsin, Sec61b (Y. Zhang, personal communication), it can be considered as superior probe for live cell imaging (Fig. 2g,i and Supplementary Table 8). PFA-fixation accelerated photobleaching of tested FPs making difference in photostability less significant compared to live cells: mBaoJin appeared to be 1.4- and 1.8-fold less photostable compared to (n1)StayGold(c4) and (n1)StayGold(c4)-E138D, respectively and 1.26-fold more photostable than mStayGold (Supplementary Fig. 20b).

Live cell imaging and super-resolution microscopy with mBaoJin

Encouraged by the high performance of mBaoJin in live mammalian cells, we sought to explore its applicability for long-term super-resolution imaging of highly dynamic subcellular structures in live cells. As imaging modalities, we chose to test spinning disk super-resolution by optical pixel reassignment and structure illumination microscopy (SIM), which are currently some of the most popular super-resolution imaging techniques for live cells. As a reference protein, we used mNeonGreen[12], which is one of the brightest and most popular monomeric GFPs. First, we confirmed that mBaoJin possessed correct localization in demanding fusions with structural proteins, including vimentin, mitochondrial presequence of human cytochrome c oxidase subunit VIII, H2B, keratin, α-tubulin, β-actin (Fig. 3a–g). Using super-resolution confocal microscopy, we could image α-tubulin and β-actin dynamics for up to 60 min without photobleaching while under identical conditions mNeonGreen lost about 20% of its initial fluorescence (Fig. 3g, h, Supplementary Video 2, 3). Furthermore, mBaoJin allowed tracking fast dynamics of microtubule-associated end-binding protein 3 (EB3) protein with a high sampling rate under continuous illumination for 2.5 min without any photobleaching (Supplementary Video 4). Using sparse deconvolution SIM[13], we achieved ~65 nm lateral resolution by imaging actin filaments and nuclear pore complex protein Nup98 in live cells (Figure 3i,j,k, Supplementary Fig. 22, 23). As expected, photobleaching rate of mBaoJin was accelerating with increasing illumination power showing similar supralinear dependence as was observed in solution (Fig. 3l, increasing power from 100 to 200 mW/mm^2 accelerated photobleaching rate by 2.8 times, see Supplementary Fig. 12), which is also typical for some other GFP-like FPs[7]. However, under all tested SIM imaging conditions with variable light intensities mBaoJin outperformed mNeonGreen and EGFP in terms of photostability (Fig. 3l, Supplementary Fig. 22, 23, Supplementary Video 5, 6). For example, under medium illumination (at 52 mW/mm^2 light power) in fusion with Lifeact, mBaoJin lost only 6% of its initial brightness in 60 s vs. 16% and 21% for EGFP and mNeonGreen, respectively (Supplementary Fig. 23b). Under higher illumination power (150-200 mW/mm^2) mBaoJin exhibited 2.5-4-fold higher photostability than mNeonGreen (Fig. 3l, Supplementary Fig. 22c). mBaoJin was also compatible with other super-resolution imaging modality, such as Bleaching/Blinking Assisted Localization Microscopy (BaLM)[14], which provides all benefits of other localization microscopy techniques, such as STORM and PALM, without the need for photoactivatable/photoswitchable fluorophores. However, mBaoJin performance in BaLM was comparable to that of mNeonGreen as both FPs demonstrated similar signal-to-noise ratios and the superior photostability of mBaoJin was not maintained at the high power densities typically applied for BaLM (Supplementary Fig. 24–27 and Supplementary Videos 7–14). Overall, our results demonstrated that mBaoJin can be considered as a superior monomeric probe for long-term super-resolution imaging of live cells under moderate and low excitation intensities (<200 mW/mm^2) typically used in confocal super-resolution and SIM imaging. Under higher excitation powers, mBaoJin can be a FP of choice when greater resistance to pH or faster maturation is needed.

To confirm that beneficial mBaoJin properties could be translated into *in vivo* and *ex vivo*, we expressed mBaoJin in neurons in *C. elegans* and mouse and quantified brightness and photostability in comparison to the selected GFPs (Fig. 4 and 5). In worms, mBaoJin exhibited ~2-fold higher intracellular brightness in neurons than mNeonGreen, enabling visualization of small processes (Fig. 4a–c). Under continuous wide-field illumination, mBaoJin demonstrated a 4-fold slower photobleaching rate compared to mNeonGreen retaining more than 50% of its initial fluorescence after 15 min of bleaching (Fig. 4d). In live mouse brain tissue, mBaoJin appeared to 1.7-fold brighter than mNeonGreen but 1.7-fold dimmer than mStayGold (Fig. 5c). However, in PFA-fixed brain tissue mStayGold and mBaoJin exhibited comparable brightness and photostability significantly outperforming mNeonGreen (Fig. 5d,e). Due to the high chemical stability of mBaoJin, it has been successfully used for expansion microscopy (ExM) to obtain super-resolution images of mitochondria and tubulin filaments in HeLa cells, and neuronal processes in mouse brain tissue (Fig. 3m–o). Importantly, mBaoJin also retained its high photostability in expanded brain tissue enabling super-resolution imaging of large volumes with less photobleaching (Supplementary Fig. 28). Thus, mBaoJin preserved its beneficial biochemical properties *in vivo* and *ex vivo*.

Discussion

To take full advantage of high brightness and photostability of StayGold for protein and membrane imaging, we performed its monomerization using unique screening approach where dimerization of the target FP controls expression of the reporter FPs (Fig. 1a,b). Fluorescence readout of reporter FPs enables high-throughput screening of large random libraries for monomeric phenotype, allowing for protein optimization without prior knowledge of the structural basis of dimerization. It was essential since when we started protein engineering, the structure of StayGold was not available. The independent effort to monomerize StayGold led by two other groups was mainly driven by structure-guided mutagenesis[5,6] and monomerization positions found in mBaoJin were different from those discovered by the structure-guided approach (Supplementary Fig. 3). In principle, this high-throughput approach can be applied to monomerization of other FPs however it still may require post-hoc validation of the selected variants with, for example, FPLC as was done in this study. Monomerization of StayGold significantly improved its intracellular brightness and accelerated chromophore maturation without affecting its high pH and chemical stability (Fig. 2d,e Supplementary Fig. 12l, and Supplementary Table 6). Despite about twice lower photostability compared to the parental StayGold and other recently introduced (pseudo)monomeric versions of StayGold, mBaoJin is still significantly more photostable than the majority of the widely used and well-validated GFPs (Fig. 2f–i, Fig. 3l,h, Fig. 4c,d, Fig. 5c,e, Supplementary Fig. 12b–k, 18, 19b, 22c, 23b, 28 and Supplementary Table6, 8) allowing for long-term (at least 60 min with photobleaching) super-resolution imaging under low and moderate light intensities. However, under extremely high illumination power

(18 W/mm^2) mBaoJin, as well as mStayGold, do not outperform other state-of-the-art GFPs, except EGFP (Supplementary Fig. 12, 24–26). High-resolution X-ray structures of mBaoJin allowed us to suggest the structural determinants of its high photostability that might be related to the denser chromophore packing by amino acids in the immediate proximity (Supplementary Fig. 11). However further photophysical studies are required to validate these suggestions. Structural analysis of mBaoJin also revealed an interesting feature: a chloride-binding pocket (Fig. 2a–c). Unlike YFP[15] and mNeonGreen[16], the presence of chloride anion in solution practically did not affect its brightness but increased its photostability (Supplementary Fig. 15–17). The mBaoJin structures, the only available structures of the StayGold monomers reported to date, will aid in the future design of enhanced and spectral versions of mBaoJin, FRET pairs, and biosensors.

mBaoJin possessed higher chemical and pH stability compared to mStayGold that translated to high performance in ExM (Fig. 2h, Fig. 3m–o). The stability of mBaoJin to fixation in PFA makes it a good marker for fluorescence microscopy of fixed tissues. Since mBaoJin, like hfYFP[10], retains its fluorescence in 6M GndHCl, it can be used to facilitate protein purification using blue light for visualization. In addition, we investigated mBaoJin for the labeling neurons in mice and worms *ex vivo* and *in vivo* (Fig. 4,5).

In conclusion, the mBaoJin protein is a bright and photostable monomeric protein with excellent behavior in fusions, which enables long-term visualization of the cytoskeleton of the cells with sub-diffraction resolution in live and fixed cells. Despite tdStayGold demonstrated superior intracellular brightness and photostability, mBaoJin presents some critical advantages over tandem dimer constructs for applications where the size of the protein matters, such as recombinant Adeno-Associated Virus (AAV) construction and biosensor engineering. We believe that mBaoJin will be a valuable probe for cell biologists and the super-resolution imaging community. In addition, the reported crystal structures may prompt further development of mBaoJin spectral derivatives and facilitate its utilization for biosensor engineering. Similarly to GFP from *Aequorea victoria* that served as a template for engineering diversity of phenotypes, we foresee generation of blue and cyan versions of mBaoJin by chromophore mutagenesis as well as other photochemical phenotypes including photoactivatable and photoswitchable GFPs with enhanced photostability compared to existing ones. Although not yet tested for mBaoJin or any other StayGold derivatives, we predict that mBaoJin will maintain its high brightness and photostability under two-photon excitation further extending its utility for *in vivo* imaging in all model organisms. Development of the circular permutated version will be crucial step towards engineering of photostable single FP-based biosensors, which can significantly extend current time scale of functional imaging.

Online Methods

Molecular cloning and mutagenesis

The StayGold gene was synthesized using polymerase chain reaction (PCR) with overlapping primers listed in Supplementary Table 9 and cloned as a BglII/EcoRI fragment in the pBAD/HisB or pBAD/HisB-Sumo vectors. For PCR amplification, we used a C1000 Touch Thermal Cycler (Bio-Rad, USA).

Random mutagenesis of StayGold gene was performed using PCR in the presence of Mn^{2+} ions in the conditions to achieve 2–3 random mutations per 1000 bp according to the Diversify PCR Random Mutagenesis Kit User Manual (Clontech, USA). The mutagenized StayGold PCR fragment was further cloned in pWA21cBP-mKate2-mTagBFP vector at PstI/BglII restriction sites to swap mKate2 gene and get StayGold in frame with $AraC_{DNA}$ gene and the library was transformed in BW25113 bacterial cells. Next, the obtained bacterial library was spread on LB/agar Petri dishes supplemented with 100 μg/ml ampicillin, 0.02% arabinose and 0.02% rhamnose and after incubation for 24 h at 37°C and 2-4 h at room temperature about twenty thousand of bacterial colonies were imaged using fluorescent stereomicroscope Leica M205FA (Leica, Germany) in blue (405/40 nm excitation and 450/40 nm emission) and green channels (480/40 nm excitation and 535/40 nm emission). About 60-70 of the brightest green and dimmer blue fluorescing colonies were further analyzed on bacterial streaks. After each round the ten brightest green/dimmer blue fluorescent variants were further purified from 5 ml of LB medium supplemented with 100 μg/ml ampicillin and 0.004% rhamnose using Ni-NTA resin and their oligomeric state was assessed using FPLC chromatography.

Directed mutagenesis of mBaoJin was performed using PCR with overlapping fragments. The PCR products after PCR with overlapping fragments were inserted into pBAD/HisB vector at BglII/EcoRI restriction sites.

Mammalian plasmid construction

In order to construct the pAAV-*CAG*-NES-GFPs-P2A-mCherry plasmids, the GFPs genes were PCR amplified as BglII-EcoRI fragments and swapped with the NCaMP7 gene in the pAAV-*CAG*-NES-NCaMP7-P2A-mCherry vector.

In order to construct the pAAV-*CAG*-mCherry-P2A-GFPs plasmids, the GFPs genes were PCR amplified as SpeI-EcoRI fragments and swapped with the EGFP gene in the pAAV-*CAG*-mCherry-P2A-EGFP vector.

In order to construct the pAAV-*CAG*-dMito-mBaoJin plasmid, the mBaoJin gene was PCR amplified as XhoI-EcoRI fragment and swapped with the mCherry gene in the pAAV-*CAG*-*d*Mito-mCherry vector.

In order to construct the pAAV-*CAG*-H2B-mBaoJin plasmid, the mBaoJin gene was PCR amplified as BglII-HindIII fragment and swapped with the B-GECO1 gene in the pAAV-*CAG*-H2B-B-GECO1 vector.

In order to construct the pLU-Vimentin-mBaoJin plasmid, the mBaoJin gene was PCR amplified as BamHI-BsrGI fragment using PCR with overlapping fragments to delete BsrGI restriction site and swapped with the NeonOxIrr gene in the pAAV-*CAG*-Vimentin-NeonOxIrr vector[17].

To construct the pEMTB-mBaoJin plasmid, the mBaoJin gene was PCR amplified as BamHI-NotI fragment and swapped with the mNeonGreen gene in the pEMTB-mNeonGreen vector (Addgene Plasmid #137802).

In order to construct the pmBaoJin-Keratin plasmid, the mBaoJin gene was PCR amplified as KpnI-NotI fragment and swapped with the mKate2 gene in the pmKate2-Keratin vector (Evrogene, Moscow, Russia).

In Piatkevich's group, synthetic DNA oligonucleotides used for cloning were synthesized by Tsingke Biotechnology Co., Ltd. or Zhejiang Youkang Biological Technology Co., Ltd., China. PrimeStar Max master mix (Takara, Japan) was used for high-fidelity PCR amplifications. Restriction endonucleases were purchased from New England BioLabs (USA) and used according to the manufacturer's protocols. DNA ligations were performed using OK Clon DNA Ligation Kit II from Accurate Biotechnology (Hunan) Co., Ltd, Changsha, China. The ligation products were chemically transformed into the TOP10 E. coli strain (Biomed, China) and cultured according to the standard protocols. Sequencing of bacterial colonies and purified plasmids were performed using Sanger sequencing (Zhejiang Youkang Biological Technology Co., Ltd., China). Small-scale isolation of plasmid DNA was performed with commercially available Mini-Prep kits (Tiangen, China); large-scale DNA plasmid purification was done with Midi-Prep kits (Qiagen, Germany). The gene of StayGold was *de novo* synthesized to substitute phiLOV3 in pAAV-CAG-phiLOV3-P2A-FusionRed plasmid by Tsingke Biotechnology Co., Ltd., China, based on the DNA sequences reported on Genbank (https://www.ncbi.nlm.nih.gov/nuccore/2204333803). The genes of mBaoJin, td8ox2StayGold, AausFP1 and mClover3 were *de novo* synthesized by Synbiob Gene Technology Co., Ltd., China. To clone pAAV-CAG-mBaoJin-P2A-FusionRed plasmid, the synthesized DNA were PCR amplified with KpnI/AgeI flanking sites and swapped with the StayGold gene in the pAAV-CAG- StayGold-P2A-FusionRed. To clone pAAV-CAG-StayGold(E138D)-P2A-FusionRed plasmid, overlap PCR with site mutation of E138D was performed with KpnI/AgeI flanking sites and swapped with the StayGold gene in the pAAV-CAG-StayGold-P2A-FusionRed. To construct pAAV-CAG-GFPs-P2A-mCherry, the genes of FusionRed in the construct of pAAV-CAG-GFPs-P2A-FusionRed were swapped with the mCherry gene flanking with SpeI/EcoRI sites. To construct CytERM (cytoplasmic end of an endoplasmic reticulum signal anchor membrane protein) fusions, StayGold, StayGold(E138D) and mBaoJin were PCR amplified with AgeI/NotI flanking sites and swapped with the mScarlet gene in the pCytERM-mScarlet-N1 (Addgene plasmid #85066). To add more control of OSER assay, EGFP, Venus, td8ox2StayGold, mClover3, mEGFP, mGreenLantern were also PCR amplified with AgeI/NotI flanking sites and swapped with the mScarlet gene in the pCytERM-mScarlet-N1. To construct plasmids for expression of structural protein fusions, mBaoJin was PCR amplified and swapped with the corresponding FP genes in pActin-Electra1 (Addgene #184941), pEB3-mScarlet-I (Addgene plasmid #98826) and pTubulin-Electra1 (Addgene #184929) plasmids. For comparison on SIM, StayGold was PCR amplified and swapped with the Electra1 gene in pTubulin-Electra1 (Addgene #184929) plasmid. For long-term super-resolution imaging on SIM, mBaoJin, EGFP, and mNeonGreen were cloned into pLifeActin-N1 plasmid.

The mammalian plasmids generated in the course of this study are available from the WeKwikGene plasmid repository at Westlake Laboratory, China (https://wekwikgene.wllsb.edu.cn/).

Protein purification and characterization

For protein expression, the genes of proteins were PCR amplified as BglII/EcoRI fragments and inserted into the pBAD/HisB (Invitrogen, USA) or pBAD/HisB-Sumo vectors at the BglII/EcoRII restriction sites, and the generated plasmids were transformed into BW25113 bacteria. The bacterial cultures were grown in 200 mL of LB medium supplemented with 0.004% arabinose and 100 μg/ml ampicillin overnight at 37°C and 180 rpm. The cultures were then centrifuged at 4648 g for 10 min. The cell pellets were resuspended in PBS buffer supplemented with 10 mM Imidazole, 300 mM NaCl, lysozyme (100 μg/ml final concentration), lysed by sonication on ice, centrifuged at 20000 rpm for 10 min, 4 °C and further purified using Ni-NTA resin (Qiagen, USA) and dialyzed for 12–16 h against PBS buffer.

For preparative mBaoJin protein purification from 2.6 L of LB medium, the BW25113 bacterial cells expressing the HisB-small ubiquitin-like modifier (SUMO)-mBaoJin protein were centrifuged for 20 min at 4648 g and 4 °C using the Avanti J-E centrifuge (Beckman Coulter, USA), cells were disrupted by sonication and the cell extract was centrifuged for 30 min at 28,000× *g*, 4 °C. The supernatant with protein was further purified using a 5 mL Ni-NTA Superflow column (Qiagen, Hilden, Germany). Next, His-SUMO-tag was cleaved using SUMO protease and the cleavage mix was applied to a Ni-NTA Superflow column (Qiagen, EU). The concentrated protein was further purified using a 1 mL ResourceQ column (GE Healthcare, Sweden) and finally concentrated until a 10 mg/mL concentration.

Extinction coefficients were determined by alkaline denaturation method as performed in *ref.* [18] by addition of the 2M NaOH water solution to protein solution in PBS buffer till 1M final concentration and assuming that GFP-like chromophore has extinction coefficient equal to 44,000 $M^{-1}cm^{-1}$ in 1 M NaOH[18]. The absorption spectra were registered using the NanoDrop 2000c spectrophotometer (Thermo Scientific, USA).

Fluorescence quantum yields were determined by the absolute method using Quantaurus-QY multichannel detector (Hamamatsu, Japan). For fluorescence measurements, the samples were diluted to have optical densities less than 0.1. Fluorescence emission and excitation spectra were measured with an LS55 spectrofluorometer (Perkin Elmer) controlled by FLWinLab™ software. The quantum yield was obtained at a set of excitation wavelengths across the absorption band nm with the step of 5 nm. In the cases where quantum yield depended on excitation wavelength because of overlapping absorption of the red (possibly anionic) and blue (possibly neutral) forms of chromophore, the quantum yield of the red form was calculated by averaging the numbers at the longest wavelength part, where the dependence reached a plateau. Each measurement was repeated 3 times and the average was accepted as final value. Quantum yields of purified proteins mBaoJin and mNeonGreen were determined in the PBS buffer by the tangent of the slope of the dependence of integral fluorescence values in the range of 480-700 nm excited at 470 nm. The EGFP protein with a quantum yield of 0.7 was used as a reference. Steady-state excitation and emission spectra were recorded using a CM2203 spectrofluorometer (Solar, Belarus) controlled by the “Universal” program software.

Fluorescence lifetimes were measured on dilute solutions with a Digital Frequency Domain system ChronosDFD (ISS) appended to a PC1 (ISS, Champaigne, IL) spectrofluorometer. The excitation of a laser (ISS) at 445 nm, selected with a 440-460 nm filter, was modulated with 30 harmonics in the range of 5–150 MHz. The fluorescence was collected at 90° through a 535/50 filter. The modulation ratio and phase delay curves were fitted to model functions corresponding to a single-exponential fluorescence decay with Vinci 3 software (ISS). Coumarin 6 in ethanol (fluorescence lifetime 2.5 ns) was used as a reference standard for obtaining instrument response function.

For determination of the maturation rates, GFPs were expressed in bacterial cells from the pBAD/HisB vector as previously described.[19] Briefly, protein expression was induced by adding 0.2% arabinose (final concentration) to 15 ml of overnight culture in a 15-ml tube filled to the brim and sealed tightly to prevent oxygen access. Protein expression lasted for 3 h at 37 °C, 220 rpm in no oxygen environment. Then cells were centrifuged at 3000 g, 10 min and the pallet was resuspended on ice in 800 μl of PBS buffer supplemented with chloramphenicol (10 μg/ml) to block protein translation and sonicated for 40 sec at 20% power using Sonics vibra cell VCX130 sonicator and CV18 tip (Sonics&Materials Inc., Newtown, CT, USA). Sonicated cells were centrifuged at 46,090 g for 2 min at 0°C. 100 μl of the supernatant was added to 2.9 ml of PBS buffer pre-warmed at 37°C and supplemented with chloramphenicol (10 μg/ml), and green fluorescence (480 ex/530 em; 5 nm slits) was recorded using a CM2203 spectrofluorometer (Solar, Belarus).

Photostabilities of purified proteins (50 μM concentration) in PBS buffer were determined in microdroplets in mineral oil. 10 μl of oil was mixed with 1.5 ul of protein solution and placed between coverslips. Protein droplets were photobleached under continuous wide-field illumination using Zeiss Axio Imager Z2 microscope (Zeiss, Germany) equipped with a X-Cite 200DC XCT200 200 W mercury arc lamp (Lumen Dynamics, Canada), a 63x 1.4 NA oil immersion objective lens (PlanApo, Zeiss, Germany), a 470/40BP excitation filter, a FT 495 beam splitter, and 525/50BP emission filter or a Nikon Ti2-E widefield fluorescence microscope equipped with Spectra III Light Engine (LumenCore, USA), 470/28BP excitation filter (Semrock), the ORCA-Flash 4.0 V3 sCMOS camera (Hamamatsu, Japan), 40x NA1.15 objective lenses (Nikon, Japan). Light power densities were measured at the focal plane of the objective lens and reported for each experiment in figure legends. The acquired photobleaching curves were presented with no corrections and with normalization by the method introduced by Shaner *et al.*[8]

For pH titrations in citric/borax buffer, 5 μl of proteins (till 50 nM final concentration) in PBS buffer were added to 200 ul of buffer (30 mM citric acid, 30 mM borax, and 30 mM NaCl) with pH values (adjusted with HCl or NaOH) ranging from 2 to 11 in 0.5 pH units interval in a 96-well black clear bottom plate (Thermo Scientific, USA). For pH titrations in citric acid/Na-citrate and monobasic/dibasic phosphate buffers, 5 μl of proteins (till 50 nM final concentration) in 50 mM sodium monobasic phosphate buffer titrated with NaOH to pH 7.20 were added to 200 ul of buffer A (100 mM citric acid titrated with NaOH to pH 2.5-6.0) and buffer B (100 mM NaH_2PO_4 titrated with NaOH to pH 5.5-9.0) with pH values ranging from 2.5 to 9.0 in 0.5 pH units interval in a 96-well black clear bottom plate (Thermo Scientific, USA). To reveal NaCl influence on pKa values, 300 mM NaCl was also added to buffers A and B. After proteins incubation for 20 min at room temperature in described buffers, the fluorescence values were measured using a ModulusTM II Microplate Reader (TurnerBiosystems, USA) equipped with fluorescence optical kit ex 490/em 510-570.

For comparison of stabilities of proteins in 6M guanidinium hydrochloride (GdnHCl), the 5 ul of proteins (till 50 nM final concentration) in PBS buffer were added to 200 ul of 30 mM HEPES, pH 7.80 buffer supplemented with 6M GdnHCl. After incubation of proteins at room temperature for 24 hours, the green fluorescence was registered using a ModulusTM II Microplate Reader (TurnerBiosystems, USA) equipped with fluorescence optical kit ex 490/em 510-570.

The oligomeric state of the proteins was characterized using ÄKTA prime plus and ÄKTA explorer 100 systems (GE Healthcare, Sweden) and a Superdex 75 10/30 GL column (GE Healthcare, Chicago, IL, USA) equilibrated with 20 mM Tris-HCl, pH 7.5, 200 mM NaCl.

Characterization in cultured mammalian cells

HEK293FT (Invitrogen), HeLa (ATCC CCL-2), and Cos7 cells (ATCC CRL-1651) cells were authenticated by the manufacturer using STR profiling, reauthenticated in our lab by inspecting stereotypical morphological features under widefield microscope and tested negative for mycoplasma contamination to their standard levels of stringency. Authentication by morphology was performed every time before transient transfection. Cos7 cells were cultured in DMEM medium supplemented with 10% fetal bovine serum in an incubator at 37 °C with 5% CO_2. HEK293FT and HeLa cells were cultured in Dulbecco's Modified Eagle Medium (DMEM) supplemented with 10% Fetal Bovine Serum (FBS) and 1% penicillin/streptomycin (PS), and seeded in 24-glass bottom well plate (P24-0-N Cellvis, USA) or glass bottom dish (MatTek, USA) after Matrigel (356235, BD Biosciences, USA) coating. Cells were transfected at 80-90% confluency using Hieff Trans Liposomal Transfection Reagent (Yeasen Biotechnology, 40802ES02) according to manufacturer's protocol, and imaged 24-48 h post transfection. Cell were imaged using a Nikon Ti2-E widefield fluorescence microscope equipped with Spectra III Light Engine (LumenCore, USA), the ORCA-Flash 4.0 V3 sCMOS camera (Hamamatsu, Japan), 10x NA0.45 and 20x NA0.75 objective lenses (Nikon, Japan) controlled by NIS-Elements AR 5.21.00 (Nikon Japan) or using a laser spinning-disk Andor XDi Technology Revolution multi-point confocal system (Andor Technology, Belfast, UK).

To measure the brightness of mBaoJin in live cells under widefield microscope, HEK cells were transfected with the P2A coexpression plasmids and were imaged in FITC (excitation 475/28 nm from SpectraIII LumenCor; emission 594/40 nm) and TRITC (excitation 555/28 nm from SpectraIII LumenCor; emission 535/46 nm) channels. To obtain statistically significant datasets, we performed 2 independent transfections and ROIs were determined using auto-detect function of NIS Elements software limiting the ROI area to 50 μm^2 as a minimal size of HEK cells. The mean fluorescence intensity in FITC and TRITC channels for ROIs was extracted, and the FITC-to-TRITC ratio were calculated after background subtraction for each channel, which was used for the comparison of intracellular brightness under corresponding imaging conditions. The data were excluded from the analysis if cells were out of focus.

To assess the brightness of mBaoJin in the cytosol of live cells under confocal microscope, HeLa cells cultured and seeded as described above were transfected with the NES-GFPs-P2A-mCherry or mCherry-P2A-GFPs plasmids using TurboFect transfection reagent (Thermo Scientific, Vilnius, Litva) and imaged in green (excitation 488 nm and emission 525/50 nm) and red (excitation 561 nm and emission 617/73 nm) channels using a laser spinning-disk Andor XDi Technology Revolution multi-point confocal system (Andor Technology, Belfast, UK) under identical imaging conditions (the same exposure times and laser power).

To measure photostability in live HEK cells, HEK cells were transfected as described above and imaged under continuous FITC wide-field excitation. A 20x NA0.75 objective lens (Nikon, Japan) and SpectraIII LumenCor were used and set at 100% power. The illumination power at the focal spot was 50.5 mW/mm^2. The photobleaching curves were calculated for each cell individually and reported as the mean photobleaching curve for each protein (averaged from all individual curves). Cells that detached or died during photobleaching experiments were excluded from data analysis.

To evaluate the photostability under live-cell confocal imaging conditions, HeLa cells were transfected with pmBaoJin-Tubulin-N1 in glass-bottomed 24-well plates and imaged for one hour after 24 h transfection using CSU-W1 SoRa imaging setup of Nikon Spinning Disk Field Scanning Confocal System with 488 nm excitation using a 40× objective lens (power at object plane 5.7 mW/mm^2).

To compare the brightness of mBaoJin with other proteins after PFA fixation, HEK cells were transfected as described above and imaged using a fluorescence wide-field microscope 36-48 h post-transfection. Cells were washed with PBS twice and fixed with 4% PFA (15714, Electron Microscopy Sciences, USA) in PBS at room temperature for 10 min. Fixed cells were gently washed with PBS twice and imaged under identical imaging settings for each protein. Image analysis was performed as described above for live cells. To obtain statistically significant datasets, 2 independent transfections were performed and ROIs were determined using the auto-detect function of NIS Elements software.

To quantify monomeric state of GFPs in mammalian cells, OSER assay[4] was utilized as described before[20]. Briefly, two sets of experiments were conducted for different expression time length whereHeLa (ATCC CCL-2) cells were cultured and transfected with the corresponding plasmids. In the first set, 3 independent experiments were performed. Cells were seeded on a 12-well glass bottom plate after Matrigel (356235, BD Biosciences) coating for 30-45 min at 37°C. Cells were then transfectedat 70-90% confluency with 1 μgDNA per well using Hieff Trans Liposomal Transfection Reagent (Yeasen; 40802ES02) according to the manufacture's protocol. Cells were imaged 12 h post-transfection using FITC channel (Nikon CSU-W1 SoRa; OBIS 488nm LX 100 mW laser; ORCA-FusionBT; 40x WI or 100xOIL; 525/50 filter). In the second set, cells were seeded on 24-well glass bottom plate after Matrigel (356235, BD Biosciences) coating and 3 independent transfections were performed with 500 ng DNA per well at 70-90 % confluency according to manufacturer's instructions (Hieff Trans Liposomal Transfection Reagent; Yeasen; 40808ES03). Cells were imaged 22 h post-transfection with FITC channel. To collect enough cells from each well, large image mode and automated image stitching were used with 10% overlap in NIS elements software. Positive cells selected for analysis had overall similar fluorescence brightness, and cells that were significantly brighter were excluded (indications of unhealthy or highly stressed cells). Cells with non-spherical nuclei, ER sheet architectures, or condensed nuclei were regarded as stressed and excluded from normal cells fraction quantification. Cell cultures with more than 25% of stressed cells were excluded from quantification of normal cells fraction.

Super-resolution BaLM imaging of the cytoskeleton of cultured mammalian cells

Immediately before imaging, the cell medium was changed to a minimal essential medium (MEM, Sigma-Aldrich) supplemented with 20 mM HEPES. Single-molecule localization super-resolution imaging of living cells was performed using Nanoimager S (ONI, UK) microscope, equipped with Olympus UPlanSApo x100 NA 1.40 oil immersion lens, 488 nm laser, 560 nm on-camera beam splitter and Scope8 sCMOS camera. Imaging was performed using three imaging condition sets: 475 W cm^{-2} 488 nm laser and 50 ms frame time (20 fps acquisition speed), 600 W cm^{-2} 488 nm laser and 10 ms frame time (100 fps acquisition speed), 1.8 kW cm^{-2} 488 nm laser, 5 ms frame time (200 fps acquisition speed). The difference between the signal-to-noise ratio of mBaoJin and mNeonGreen localizations was tested using the Kolmogorov–Smirnov test.

Image acquisition and super-resolution reconstruction were performed using NimOS 1.18.3.15066 (ONI, UK). Image reconstruction was performed using default parameters. Data analysis was performed using FiJi ImageJ 1.53f51 (ref.[21]) and custom Python 3.9 scripts. For blinking duration calculation, the "Tracking" tool of NimOS was used. Spatial resolution was calculated using decorrelation analysis[22] with default parameters. As a widefield image, the standard deviation of images stack was used.

Super-resolution HIS-SIM imaging of nuclearpores and actin filaments in Cos7 cells

The procedure for imaging was performed by following the previous report[23]. Briefly, Cos7 cells, cultured as described above, were seeded in in 12 well plate and transfected using Lipofectamine 2000 (ThermoFisher Scientific) according to the manufacturer's instructions with 500 ng of the EGFP-Lifeact/mNeonGreen-Lifeact/mStaygold-Lifeact/ mNeonGreen-Nup98/ mStaygold-Nup98 plasmid per well. After 24 hours of transfection, cells were replated on coverslips (H-LAF 10L glass; reflection index, 1.788; diameter, 26 mm; thickness, 0.15 mm, customized) coated with 10% poly-l-lysine solution (Sigma) for ~24 h before seeding transfected cells. Live cells were imaged in HBSS (Gibco) at 37°C in a humidified live cell imaging workstation for live SIM imaging. Super-resolution imaging of nuclearpores and actin filaments were performed using commercialized HIS-SIM (High Intelligent and Sensitive Structured Illumination Microscope) provided by Guangzhou Computational Super-resolution Biotech Co., Ltd. Images were acquired using a 100×/1.5 NA oil immersion objective (Olympus) using 488 nm laser. To further improve the resolution and contrast in reconstructed images, sparse deconvolution was used by the previous report[13]. The software of HIS-SIM, IMAGER, was used for data collection and export. Data analysis was performed using Fiji 2.9.01/1.53t ImageJ.

Characterization of GFPs in mouse brain tissue

All animal maintenance and experimental procedures for mice were conducted according to the Westlake University Animal care guidelines, and all animal studies were approved by the IACUC of Westlake University, Hangzhou, China under animal protocol #19-044-KP-2. Procedures involving experimental animals are reported in accordance with Animal Research: Reporting of In Vivo Experiments (ARRIVE). Mice were maintained at strict barrier facilities with macroenvironmental temperature and humidity ranges of 20–26 °C and 40–70%, respectively. Food and water were provided ad libitum. The rooms had a 12 h light–12 h dark cycle. The housing conditions were closely monitored and controlled. C57BL/6 mice supplied by the Animal Facility at Westlake University were used without regard for sex. For expression of GFPs in the brain, the neonatal intraventricular injections of custom-made recombinant AAVs, serotype 2/9 (rAAV2/9-CAG-GFPs-P2A-FusionRed, titer: >10^{12} v.g. ml^{-1}; Shanghai Sunbio Medical Biotechnology,China) as described previously[20]. In brief, viruses were injected pan-cortically into pups at postnatal day 0 regardless of sex with a Hamilton microliter syringe. For each hemisphere, 0.5 µl virus solution (titer: >10^{12} v.g. ml^{-1}) supplemented with 0.1% FastGreen dye (Sigma-Aldrich) was injected under the skull manually. After injections in both hemispheres, the pups were returned to the home cages. The whole injection process for each pup was controlled within a few minutes to prevent cannibalism. For assessment of intracellular brightness, acute brain slices were obtained from 7-week old mice and at least 10 FOVs were imaged using Nikon wide-field microscope described above. To assess the performance of GFPs in fixed tissue and ExM, mice were deeply anesthetized and perfused using 4% PFA and at least two 50-µm brain slices from two mice at postnatal days 50 were imaged and processed. Fixed brain tissue samples were imaged using confocal microscope Zeiss LSM 980 (Germany; 10x NA 0.45, 40x NA 0.95) controlled by ZEN blue v3.5. For measuring the photostability of GFPs in fixed brain slice, two brain slices from two mice for each protein were imaged under continuous FITC wide-field excitation. A 20x NA0.75 objective lens (Nikon, Japan) and SpectraIII LumenCor were used and set at 50% power. The illumination power at the focal spot was 25.9 mW/mm^2. The photobleaching curves were calculated for each neuron individually and reported as the mean photobleaching curve for each protein (averaged from all individual curves).

Characterization of GFPs in *C. elegans* neurons

The FPs were expressed using extrachromosomal arrays in *C. elegans* and assessed for intracellular localization, brightness and photostability in the green channel. For expression in *C. elegans*, the target genes were codon-optimized using *C. elegans* Codon Adapter application (https://worm.mpi-cbg.de/codons/cgi-bin/optimize.py) with insertion of two introns[24] and *de novo* synthesized (Tsingke Biotechnology Co., Ltd, China). The codon optimized genes of GFPs were cloned into pSF11 vector (Addgene plasmid #179485) under the pan-neuronal *tag-168* promoter. The establishment of transgenic lines was done by SunyBiotech Co. Ltd (China) according to standard protocols. Briefly, wild-type N2 worms were co-injected with two plasmids encoding wGFPs::wmTagBFP2 with 10 ng/µl final concentration each. wmScarlet (Psur-5::sur-5::NLSwmScarlet) was used as a reporter marker for screening positive worms (provided by SunnyBiotech Co. Ltd, China). Transgenic lines were selected by green fluorescence and confirmed with sequencing of the target. Selected worms were maintained and grown on nematode growth medium (NGM) plates seeded with E. coli OP50-1 at 20°C following standard protocols. Positive worms with green fluorescence (used without regard to sex) at L4 stage of development were selected for further imaging. Worms were mounted on immunofluorescence assay slides (Jiangsu Shitai Zhenduan Jishu Co. Ltd, 80383-0209-01), immobilized with 25 mM levamisole and imaged using a wide-field Nikon microscope with 10x NA0.45 objective lens for quantification and confocal microscope Zeiss LSM 980 for structural imaging (Germany; 10x NA 0.45, 40x NA 0.95) controlled by ZEN blue v3.5. Individual neurons were selected manually, and mean fluorescence intensity was calculated using Nikon Elements software.

Expansion Microscopy of HEK cells and brain tissue

HEK293FT and HeLa cells, cultured as described above, were seeded onto Matrigel treated (356235, BD Biosciences, USA) glass-bottom dishes (MatTek, USA) and transfected at 40-50% confluency with the pAAV-CAG-GFPs-P2A-mChilada plasmids (mChilada is a new RFP described in the upcoming pre-print, gift from Nathan Shaner) and mixture of pN1-mBaoJin-mito and pN1-mTurquise2-H2B plasmids, respectively, using Lipofectamine 3000 (Thermo Fisher Scientific, USA) according to the manufacturer's protocol. After 36-48 hours, cells were briefly rinsed with 1x PBS and fixed for 10 min with 4% PFA (Vendor) and imaged before undergoing ExM procedure.

ExM was performed according to the optimized protocol called Octa-ExM[25]. Briefly, PFA-fixed mice brain slices and cultured mammalian cells were treated with 5% allyl glycidyl ether (TCI, Japan) dissolved in 100 mM $NaHCO_3$ for 2 hours at 37°C, followed by 3 brief washes with 1xPBS. The Stock8X monomer solution (8.624% (w/v) sodium methacrylate, 31.37% (w/v) N,N-dimethylacrylamide, 0.385% (v/w) trimethylolpropane propoxylate triacrylate, all from Sigma-Aldrich, United States) was mixed in water, adjusted to pH=6.5 with HCl, and stored at 4°C before use. Activated monomer solution were prepared by adding freshly prepared 10% (w/v) solution of ammonium persulfate (APS) initiator and tetramethyl-ethylenediamine (TEMED) accelerator (Sigma-Aldrich, United States) into Stock8x monomer solution to final concentration of 0.3% (w/v) and 0.2% (w/v), respectively. Pre-treated samples were place into glass-bottom dish (MatTek, USA) and incubated in activated Stock8x monomer solution for 30 minutes for brain sections and 10 minutes for cell culture at 4°C. Gelation was carried out in an anaerobic and humid environment for 3 hours for brain sections and for 2 hours for cell cultures at 37°C. The fully formed gel were then submerged with homogenization buffer containing 8 U/ml proteinase K (NEB, United States or YEASON, China). Sample were homogenized overnight at RT (brain sections) or for 4 hours at 37°C (cell cultures). Homogenized sample-hydrogel composites were expanded in doubly deionized water for 4-5 h and imaged.

Protein crystallization

An initial crystallization screening of mBaoJin was performed with a robotic crystallization system (Rigaku, Woodlands, TX, USA) and commercially available 96-well crystallization screens (Hampton Research, Aliso Viejo, CA, USA and Anatrace, Maumee, OH, USA) at 15°C using the sitting drop vapor diffusion method. The protein concentration was 8.3 mg/mL in the following buffer: 50 mM Tris, 150 mM NaCl pH 7.5. The initial conditions were optimized by the sitting-drop vapor-diffusion method in 24-4 Intelli plates. The crystals were obtained within several days under the following conditions: 28% PEG3350, 0.2 M lithium sulfate, 0.1 M sodium acetate pH 4.6; 30% PEG8000, 0.1M ammonium acetate, 0.1M sodium cocadilate pH 6.5, and 27% PEG 3350, 0.2M lithium sulfate, 0.1 M Tris-Cl pH 8.5.

Data collection, structure solution, and refinement

Crystals were briefly soaked in a cryosolution containing precipitant supplemented with 20% Glycerol (Hampton Research, Aliso Viejo, CA, USA) immediately prior to diffraction data collection and flash-frozen in liquid nitrogen. The X-ray data were collected using 0.97861 Å wavelength from a single crystal (pH 6.5) at 100 K at beamline BL19U1 of the National Facility for Protein Science Shanghai at Shanghai Synchrotron (Shanghai, China). The X-ray data for other crystals (pH 4.6 and 8.5) were collected using 1.54 Å wavelength at 150K at Regaku XtalLAB Synergy-S laboratory system (The Woodlands, Texas, USA). The data were indexed, integrated, and scaled using the XDS program[26] (Supplementary Table 10). Structure of mBaoJin crystallized at pH 6.5 (PDB ID: 8Q79) was solved using MOLREP program[27] with a subunit A of *pdb* 8BXT as a model[6] [https://www.rcsb.org/structure/8BXT]. For other structures (PDB ID: 8QBJ, 8QDD) mBaoJin-pH6.5 was used as a model. Refinement was carried out using the REFMAC5 program of the CCP4 suite[28]. The visual inspection of electron density maps and the manual rebuilding of the model were carried out using the COOT interactive graphics program[29]. The hydrogen atoms in fixed positions were introduced during the refinement. In the final model, an asymmetric unit of every structure contained two independent copies of the protein with chromophores and solvent molecules. The mBaoJin structures were deposited in the Protein Data Bank[30] with PDB ID accession codes 8QBJ (pH4.6), 8Q79 (pH 6.5) and 8QDD (pH 8.5). Ramachandran statistics showed 98% (mBaoJin pH4.6), 99 % (mBaoJin pH6.5) and 98% (mBaoJin pH8.5) of residues in the most favored region and 2% (mBaoJin pH4.6), 1% (mBaoJin pH6.5), and 2% (mBaoJin pH8.5) in the allowed regions.

Data analysis, image processing, statistics, and data visualization

To estimate the significance of the difference between two values, we used the Mann–Whitney Rank Sum Test and provided p-values calculated for the two-tailed hypothesis. We considered the difference as significant if the p value was < 0.05. The mean fluorescence intensity in the green and red channels for ROIs corresponding to the cytosol region was extracted, and after background subtraction for each channel, the green-to-red ratio was calculated and used to compare intracellular brightness.

Fluorescence brightness and photostability data were analyzed offline using NIS-Elements Advance Research software (v5.21.00; Nikon Japan), Origin (OriginLab 2019b), Excel (Microsoft Office Professional Plus 2021). Cells were selected automatically using built in function of NIS-Elements and background regions were selected manually and fluorescence measurements were performed for each region of interest (ROI), and then fluorescence from an background region was subtracted from cell fluorescence to correct for background, normalized in Excel and ploted in Origin. The raw data for 2D-SIM images were acquired using Hessian-SIM, and subsequent image reconstruction was performed using Wiener deconvolution in HiS-SIM IMAGER software (version v1.2.3.d). The Wiener deconvolution utilized North filter with a 30-pixel setting and Defocus elimination set to 0.5. Sparse-SIM, as an additional processing step, involved Sparse deconvolution based on the 2D-SIM results. This step was carried out using MicroscopeX FINER software (version v1.0.9b). For actin, the sparse parameters included sample type (-actin), sparse level (level1), and specific parameters for Nuclearpore, such as sparse iteration times (200), image fidelity (800), t axial continuity (0.1), sparsity (10), deblurring method (Landweber), and oversampling method (Fourier Sampling).

Boxplots with notches are used in Fig. 2, 4, 5: narrow part of notch, median; top and bottom of the notch, 95% confidence interval for the median; top and bottom horizontal lines, 25% and 75% percentiles for the data; whiskers extend 1.5 × the interquartile range from the 25th and 75th percentiles; horizontal line, mean; outliers not shown but included in all calculations and available in the source datasets.

Supplementary Material

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Acknowledgments

We are grateful to Aisylu N. Asanova and Anastasiya N. Romantsova for their help with mammalian plasmids cloning. We thank Gerard G Lambert and Nathan Shaner for providing the mChilada gene. We thank Guangzhou CSR Biotech Co. Ltd assistance with live-cell imaging and image analysis using HIS-SIM. This research was partially funded by an grant of the National Research Center Kurchatov Institute (development of mBaoJin and characterization of mBaoJin in vitro) and thematic plan of NRC Kurchatov Institute (crystallization and X-ray analysis); by the Ministry of Science and Higher Education of the Russian Federation for the development of the Kurchatov Center for Genome Research 075-15-2019-1659 (protein purification); by start-up funding from the Foundation of Westlake University, Westlake Laboratory of Life Sciences and Biomedicine, National Natural Science Foundation of China grant 32171093 and "Pioneer" and "Leading Goose" R&D Program of Zhejiang 2024SSYS0031 to K.D.P (characterization of mBaoJin in mammalian cells and super-resolution imaging). Assessment of the mBaoJin reporter in BaLM super-resolution microscopy was funded by RSF, project number 22-14-00400, https://rscf.ru/project/22-14-00400/. The work was also supported by the Resource Centers department of the National Research Center Kurchatov Institute (imaging of bacteria). The authors acknowledge the BL19U1 beamline of National Facility for Protein Science Shanghai at Shanghai Synchrotron.

Data availability

The entry IDs in the Protein Data Bank are 8QBJ, 8Q79, and 8QDD for atomic structures of mBaoJin crystallized at pH 4.6, pH 6.5, and pH 8.5, respectively. The total size of the files acquired for this study exceeds the 20 GB limit of the FigShare repository, therefore only the most essential raw datasets including source files for Supplementary Figures and raw unprocessed images, are available at FigShare (10.6084/m9.figshare.25001768). The rest of the files are available from the corresponding author upon request. All plasmids used in this study are available from WeKwikGene (https://wekwikgene.wllsb.edu.cn/). Source data files are provided with this paper.

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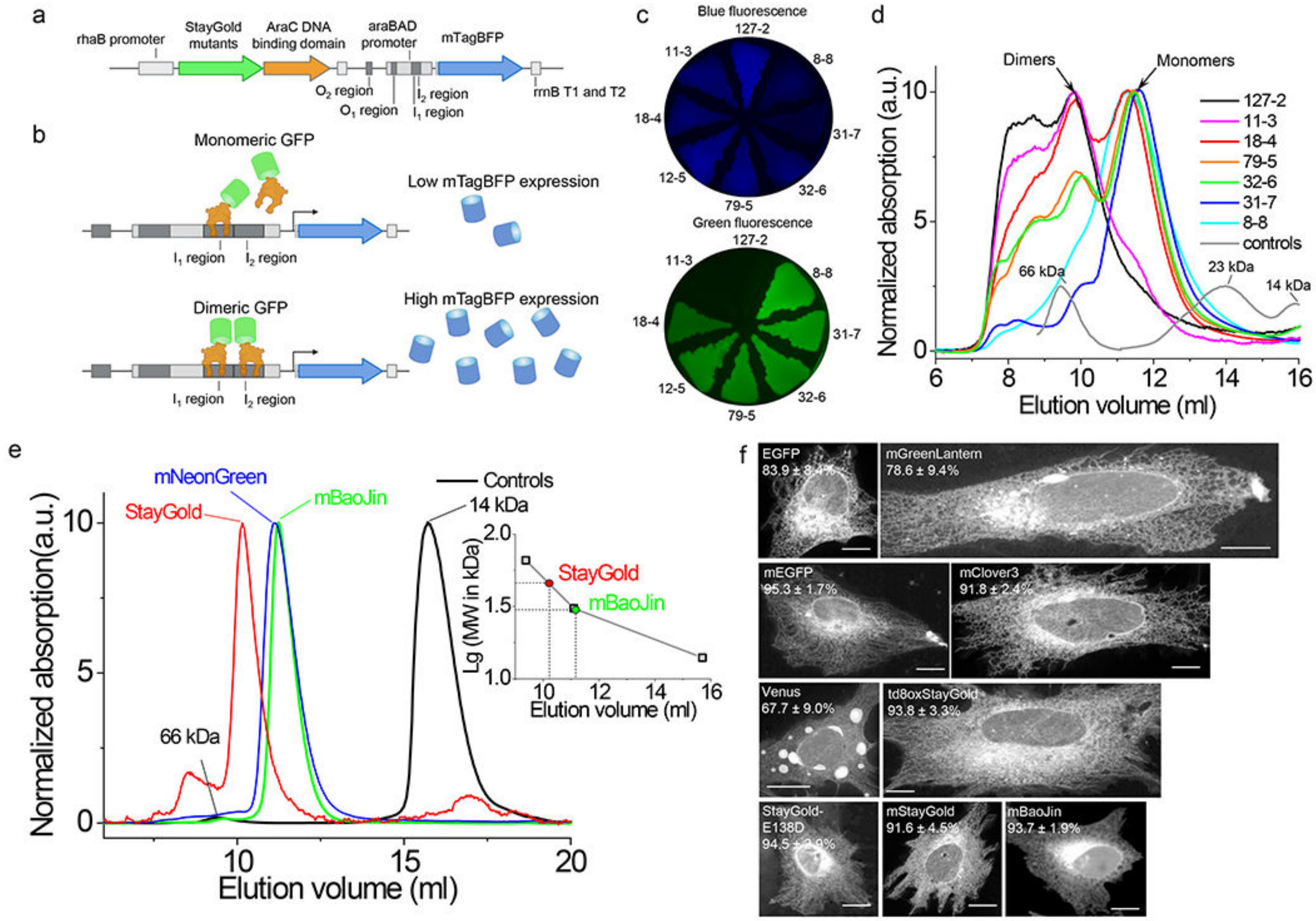

Figure 1. Monomerization of the StayGold protein.

(a) Linear map of expression cassettes in the pWA21cBP vector design for screening of monomeric StayGold variants. (b) Proposed model for reporter gene expression regulation using the engineered vector shown in a. (c) Fluorescence images of bacterial streaks expressing the selected mutants using the pWA21cBP vector in blue and green channels. (d) FPLC chromatograms for the variants shown in panel c. BSA (66 kDa), papain (23 kDa), and lysozyme (14 kDa) were used as molecular weight standards. (e) FPLC chromatograms of the mBaoJin, mNeonGreen, and original StayGold proteins. Insert shows calibration curve for molecular mass calculation. BSA (66 kDa), mNeonGreen (31 kDa), and lysozyme (14 kDa) were used as molecular weight standards. (f) Representative images of live HeLa cells expressing CytERM fusions of EGFP, mGreenLantern, mEGFP, mClover3, Venus, td8oxStayGold, StayGold-E138D, mStayGold and mBaoJin (n=1497, 1744, 2569, 1383, 1585, 1478, 2211, 2107, 1744 cells from three independent transfections from 3 independent cultures each, respectively, image 12 h post-transfection; values indicate fraction of normal cells ± SD (see Supplementary Table 1 for complete descriptive statistics); dynamic range of all images was adjusted independently to facilitate visualization of fine structures). Scale bars, 10 μm.

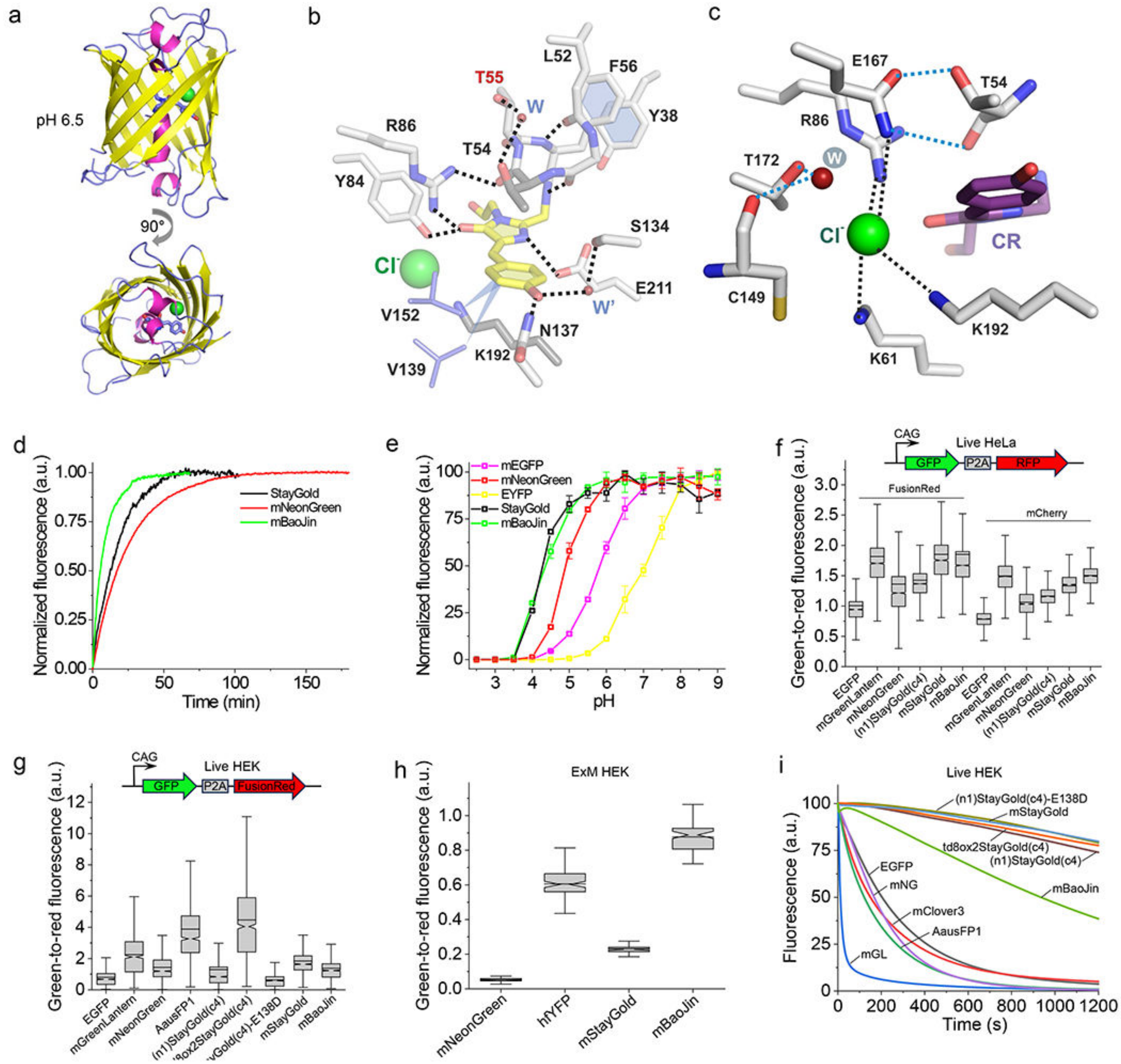

Figure 2. Structure and biochemical properties of purified mBaoJin and its behavior in mammalian cells.

(a) Overall 3D structure of mBaoJin at pH 6.5 (green sphere, chloride anion). (b) Immediate chromophore environment at pH 6.5. (c) Chloride binding pocket in mBaoJin at pH 6.5. (d) Fluorescence maturation kinetics in solution at 37°C (n = 4, 4, and 3 biological replicates for StayGold, mBaoJin, and mNeonGreen, respectively; solid line, averaged curve; fluorescence was normalized to the maximum intensity value of corresponding FP). (e) Fluorescence pH stability measured in the presence of 300 mM NaCl (n = 3 technical replicates each; squares, mean; error bars, SD; fluorescence was normalized to the maximum intensity value of corresponding FP). (f) Normalized intracellular brightness in live HeLa cells (n =

3427, 3353, 3066, 3064, 2196, 3305, 3304, 3722, 3310, 3718, 2234, 3235 cells for EGFP, mGreenLantern, mNeonGreen, (n1)StayGold(c4), mStayGold, mBaoJin with FusionRed and mCherry respectively, from 8 independent transfections from 4 independent cultures each except for mStayGold (4 independent transfections from 2 independent cultures); top of the panel, schematics of the used expression cassette; brightness for each FP was normalized to the FusionRed or mCherry signal; expression time 45-46 h). Box plots with notches used in panels f, g, and h (see Methods for description). (g) Normalized intracellular brightness in live HEK cells imaged using standard GFP filter (n = 1390, 1076, 1658, 1216, 1475, 1290, 1354, 1301, and 1604 cells for EGFP, mGreenLantern, mNeonGreen, AausFP1, (n1)StayGold(c4), td8oxStayGold, (n1)StayGold(c4)-E138D, mStayGold, mBaoJin from four independent transfections from two independent cultures each; schematic of the used expression cassette (top); brightness for each FP was normalized to the FusionRed signal; expression time 23-26 h). (h) Green-to-red fluorescence of ExM treated HEK cells co-expressing GFPs with mChilada RFP (n = 85, 87, 81, 81 cells for mNeonGreen, hfYFP, mStayGold, mBaoJin, respectively, from one independent transfection each). (i) Photobleaching curves in live HEK cells under wide-field illumination (n=80 cells from 4 independent transfections from 2 independent cultures each; fluorescence was normalized to the intensity value of corresponding FP at t = 0 s and was not corrected for emission rates).

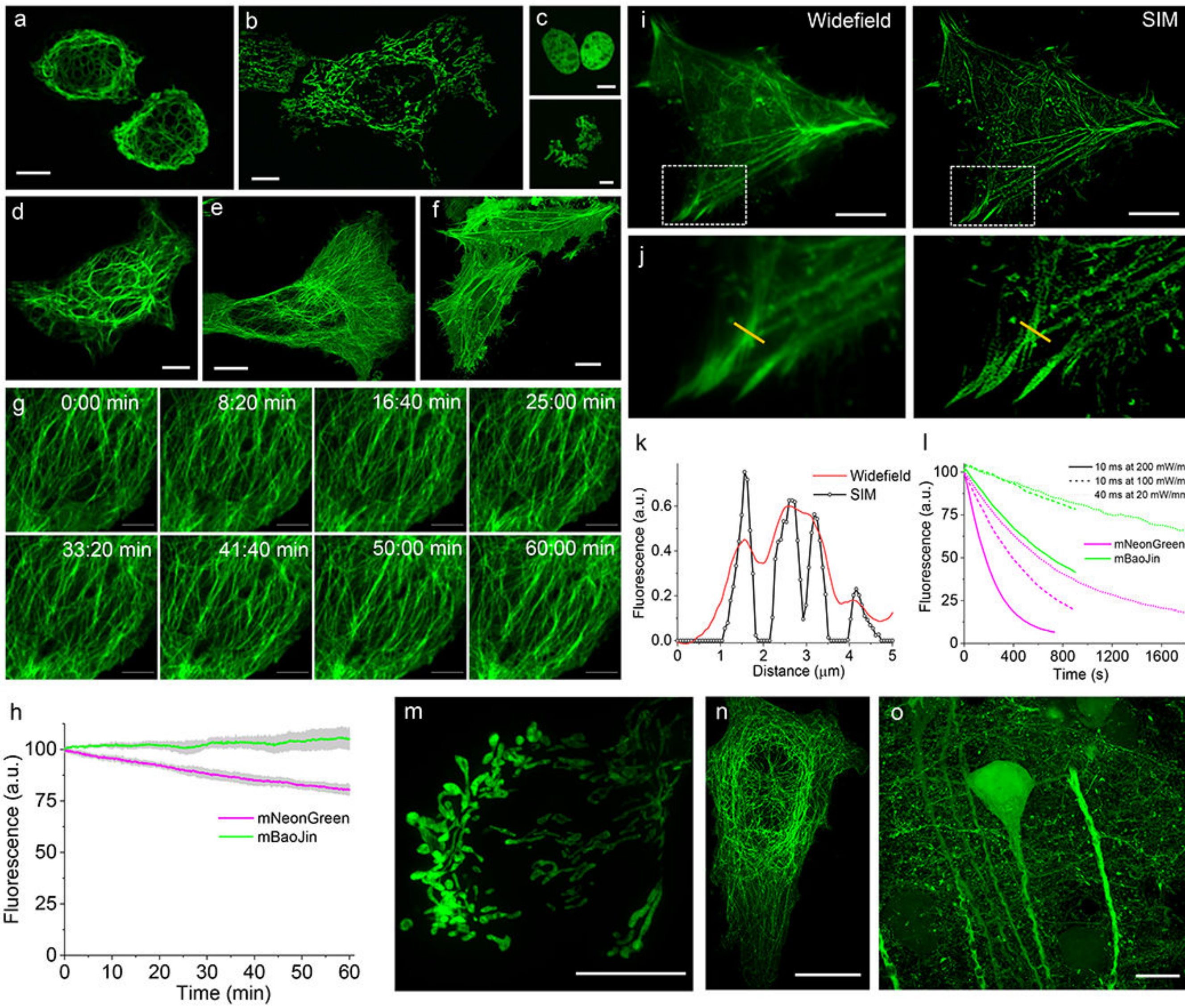

Figure 3.

mBaoJin enables long-term super-resolution imaging of live HeLa cells. (a-f) Confocal images of HeLa cells expressing mBaoJin fusions with (a) vimentin, (b) mitochondrial presequence of human cytochrome c oxidase subunit VIII, (c) H2B (lower panel shows mitotic nuclei), (d) keratin, (e) α-tubulin, (f) β-actin (n= 10, 9, 12, 4, 55, 23 cells from 4, 3, 2, 2, 4, and 2 independent transfections, respectively). Scale bars, 10 μm. (g) Long-term confocal super-resolution imaging of tubulin dynamics using CSU-W1 SoRa imaging setup at 16 mW/mm^2 (n = 32 cells from 6 independent transfections from 2 cultures; imaging frequencies 0.2 Hz). Scale bars, 1 μm. (h) Photobleaching curves of tubulin microtubules during the experiment shown in g (n = 28, 32 cells from 2, 6 independent transfections from 1, 2 cultures for mNeonGreen and mBaoJin, respectively; solid line, mean; shaded area, standard deviation; fluorescence was normalized to the intensity value of corresponding FP at t = 0 s). (i) Representative images of HeLa cell expressing mBaoJin-actin under widefield (left) and SIM (right) microscopy (n=4 cells from 2 independent transfection).

Scale bars, 10 µm. (j) Magnified views of boxed regions in i. (k) Quantification of line cuts in j. (l) Photobleaching curves for mBaoJin and mNeonGreen during SIM imaging of live cells at indicated light powers (n=3 cells for each condition for each protein from 2 independent transfections; fluorescence was normalized to the intensity value of corresponding FP at t = 0 s and was not corrected for molecular brightness and excitation spectrum). (m-o) ExM images of mBaoJin localized in mitochondria (m; n = 25 cells from four independent transfections from four independent cultures) and tubulin (n; n = 10 cells from two independent transfections from one culture) in HeLa cells and in neurons (o; n = 2 slices from two mice) in the mouse brain tissue. Scale bars, 5 µm (linear expansion factor 3.5x), 10 µm (linear expansion factor 4.2x), 20 µm (linear expansion factor 5.3x).

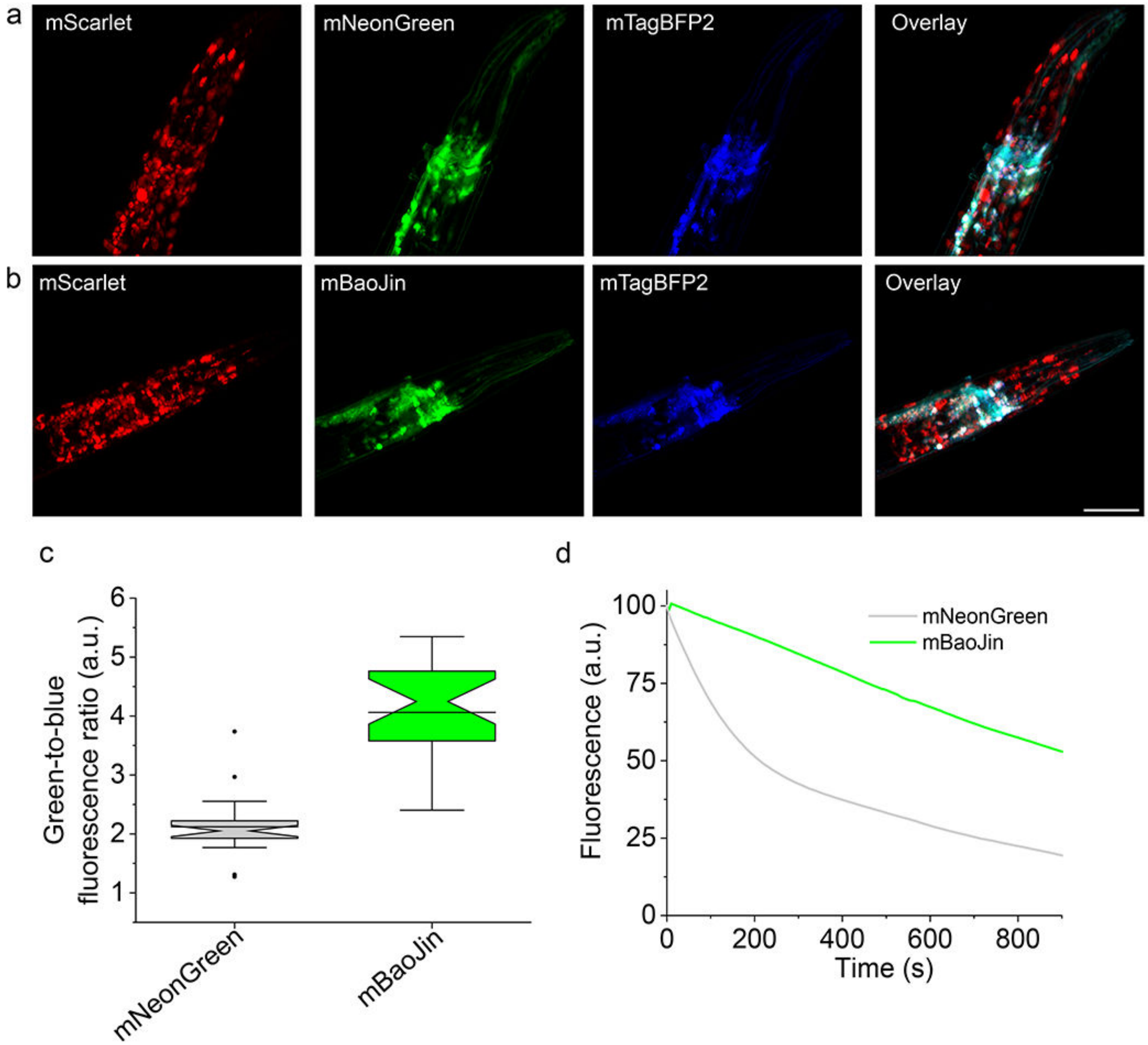

Figure 4.

Characterization of mNeonGreen and mBaoJin in neurons in live *C. elegans*. (a) Representative fluorescence images of live *C. elengans* co-expressing mNeonGreen and mTagBFP2 in neurons and mScarlet in the nucleus of muscle cells (n=5 worms from one transgenic line). Scale bar, 25 μm. (b) Representative fluorescence images of live *C. elengans* co-expressing mBaoJin and mTagBFP2 in neurons and mScarlet in nucleus of muscle cells (n=5 worms from one transgenic line). Scale bar, 25 μm. Dynamic range for images in a and b was adjusted independently to facilitate visualization and images were generated using maximum projection of z-stack acquired under confocal microscope. For brightness quantification see panel c. (c) Quantification of intracellular brightness of mNeonGreen and mBaoJin perfromed by normalization to mTagBFP2 brightness imaged under wide-field microscope (n=25 neurons from 5 worms each). Box plots with notches: narrow part of notch, median; top and bottom of the notch, 95% confidence interval for the median; top and bottom horizontal lines, 25% and 75% percentiles for the data; whiskers extend 1.5 × the interquartile range from the 25th and 75th percentiles; horizontal line, mean; outliers not shown but included in all calculations and available in the source datasets. (d) Photostability curves for mNeonGreen and mBaoJin under continuous wide-field illumination at 475/25 nm of 68 mW/mm^2 (fluorescence was normalized to the intensity value of corresponding FP at t = 0 s and was not corrected for molecular brightness and excitation spectrum).

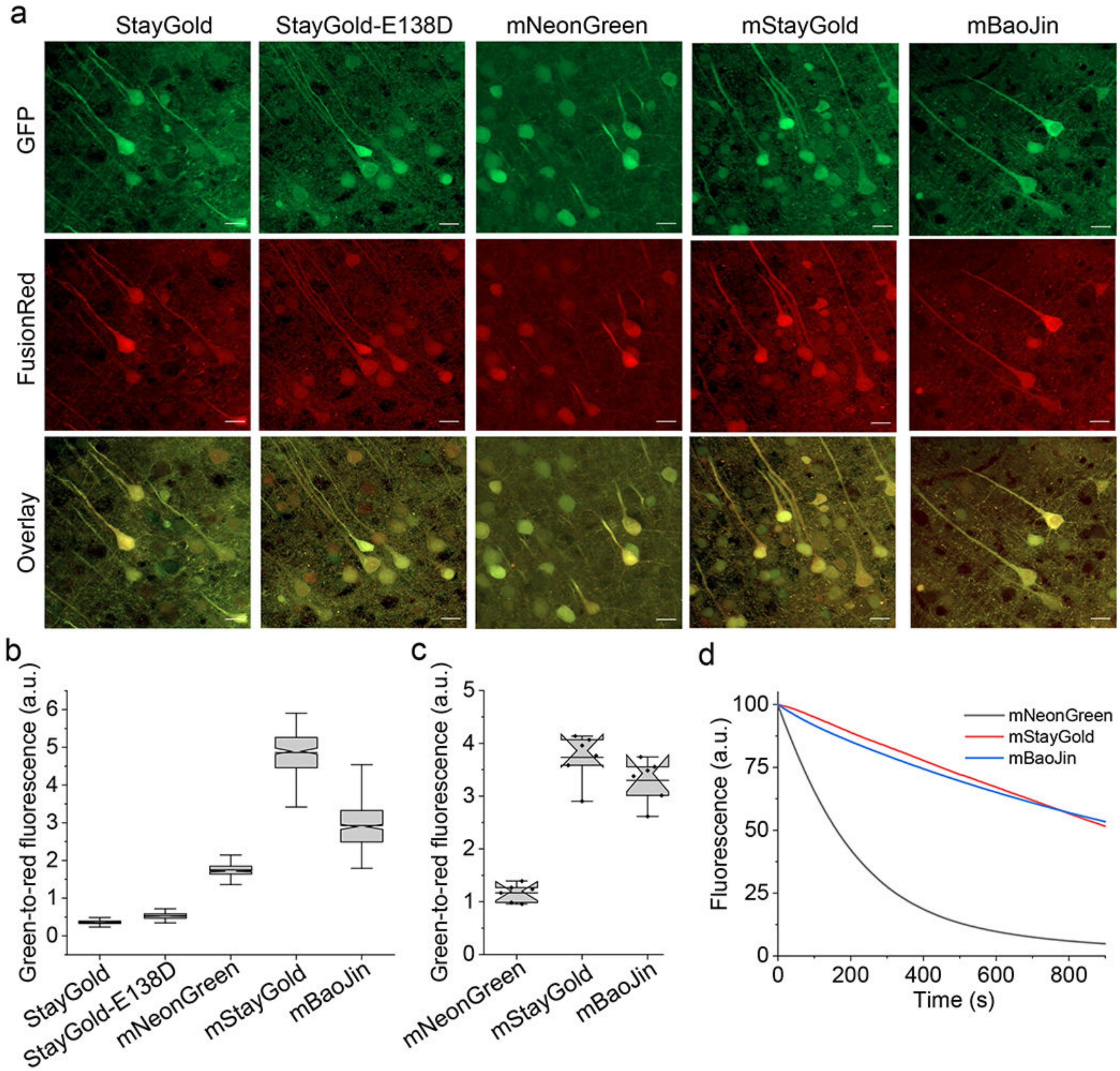

Figure 5.

Characterization of the selected GFPs expressed in L2/3 cortical neurons in mouse brain tissue. (a) Representative confocal fluorescence images of fixed brain slices expressing GFPs-P2A-FusionRed (n = 4 slices from two mice for each protein). Imaging conditions, GFP: excitation 488 nm, emission 500–550 nm; FusionRed: excitation 561 nm, emission 580-620 nm. To facilitate visual comparison of FP localization, the dynamic range was adjusted independently for each image and images were generated through maximum projection. Scale bars, 20 μm. (b) Intracellular normalized brightness of GFPs imaged in acute brain slices from 7-week-old mice (n = 284, 403, 398, 249, 397 neurons for StayGold, StayGold-E138D, mNeonGreen, mStayGold, mBaoJin from 3 slices one mice for each protein). Brightness for each FP was normalized to the FusionRed signal (imaging conditions same as in a). Box plots with notches used in panels c and d: narrow part of notch, median; top and bottom of the notch, 95% confidence interval for the median; top and bottom horizontal lines, 25% and 75% percentiles for the data; diamonds, individual data points; whiskers extend 1.5 × the interquartile range from the 25th and 75th percentiles; horizontal line, mean; outliers not shown but included in all calculations and available in the source datasets. (c) Intracellular normalized brightness of GFPs imaged in PFA-fixed brain slices from 7-week-old mice (n = 6 field of views from two slices from two mice for each protein). Brightness for each FP was normalized to the FusionRed signal (imaging conditions same as in a). (d) Normalized photobleaching curves of GFPs measured in PFA-fixed brain slices (n = 4 field of views from two slices for two mice for each protein; fluorescence was normalized to the intensity value of corresponding FP at t = 0 s and was not corrected for molecular brightness and excitation spectrum). Imaging conditions, excitation 475/28 nm at 25.9 mW mm^{-2}, emission 535/46 nm.